

Practice Problems

Question 1

Your friend is deciding between continuing to use a gasoline scooter or switching to an electric bike for their daily commute. The scooter consumes fuel at a rate of 35 km/l, which runs on gasoline, and the energy content of gasoline remains 10 kWh/l. The electric bike uses 0.05 kW h/km. Which mode of transport uses less energy daily and by how much, considering that the daily commute is 25 km %?

- Calculate the total energy consumption for the electric bike.

$$E_{\text{e-bike}} = d \cdot e_{\text{e-bike}} \quad (1)$$

$$= 25 \cdot 0.05 \quad (2)$$

$$= 1.25 \text{ kWh} \quad (3)$$

- Calculate the total energy consumption for the gasoline scooter.

$$\text{Fuel used} = \frac{25}{35} \quad (4)$$

$$= 0.714 \text{ l} \quad (5)$$

$$E_{\text{scooter}} = 0.714 \cdot 10 \quad (6)$$

$$= 7.14 \text{ kWh} \quad (7)$$

- Which mode of transport uses less energy and by how much percentage?

$$\text{Energy savings} = \frac{7.14 - 1.25}{7.14} \quad (8)$$

$$= 0.825 \quad (9)$$

$$\text{Percentage savings} = 0.825 \cdot 100 \quad (10)$$

$$= 82.5\% \quad (11)$$

Therefore, the electric bike uses 82.5% less energy than the gasoline scooter over the same distance.

Question 2

Two friends are comparing the energy efficiency of their vehicles for a 100 km weekend trip:

- Sam drives an electric vehicle (EV) that consumes 0.18 kWh/km.
- Alexa drives a hydrogen car that consumes 1.1 kg of hydrogen per 100 km.
- The energy content of hydrogen is 33.3 kWh/kg.

(a) Calculate the total energy used by each vehicle for the 100 km trip.

$$E_{\text{EV}} = 0.18 \frac{\text{kWh}}{\text{km}} \cdot 100 \text{ km} \quad (12)$$

$$= 18 \text{ kWh} \quad (13)$$

$$E_{\text{Hydrogen}} = 1.1 \text{ kg} \cdot 33.3 \frac{\text{kWh}}{\text{kg}} \quad (14)$$

$$= 36.63 \text{ kWh} \quad (15)$$

(b) Which vehicle is more energy-efficient, and by how many percent?

$$\text{Energy savings} = 36.63 \text{ kWh} - 18 \text{ kWh} \quad (16)$$

$$= 18.63 \text{ kWh} \quad (17)$$

$$\text{Percent difference} = \frac{18.63}{36.63} \cdot 100 \quad (18)$$

$$\approx 50.9\% \quad (19)$$

Question 3

A regional transport agency is evaluating whether to replace a short-haul domestic flight route (300 km) between two coastal cities in New Zealand with a high-speed electric ferry. Each mode carries 200 passengers per trip.

You are given the following information:

Parameter	Domestic Aircraft	High-speed Ferry
Energy consumption (kWh/km)	20	15
Passenger capacity	200	200
Distance	300 km	300 km

- Distance per trip: $d = 300$ km
- Passenger count: $N = 200$
- Aircraft energy consumption: $e_{\text{air}} = 20$ kWh/km
- Ferry energy consumption: $e_{\text{ferry}} = 15$ kWh/km

(a) Calculate the energy consumption per passenger-km for both the aircraft and the ferry.

Aircraft

$$\text{Total energy} = e_{\text{air}} \cdot d \quad (20)$$

$$= 20 \cdot 300 \quad (21)$$

$$= 6000 \text{ kWh} \quad (22)$$

$$\text{Energy per passenger-km} = \frac{6000}{200 \cdot 300} \quad (23)$$

$$= 0.1 \frac{\text{kWh}}{\text{pax} \cdot \text{km}} \quad (24)$$

Ferry

$$\text{Total energy} = e_{\text{ferry}} \cdot d \quad (25)$$

$$= 15 \cdot 300 \quad (26)$$

$$= 4500 \text{ kWh} \quad (27)$$

$$\text{Energy per passenger-km} = \frac{4500}{200 \cdot 300} \quad (28)$$

$$= 0.075 \frac{\text{kWh}}{\text{pax} \cdot \text{km}} \quad (29)$$

(b) Determine the total energy required per trip for each mode.

$$E_{\text{aircraft}} = 6000 \text{ kWh} \quad (30)$$

$$E_{\text{ferry}} = 4500 \text{ kWh} \quad (31)$$

(c) How much energy is saved per trip by switching to the ferry?

$$\Delta E = E_{\text{aircraft}} - E_{\text{ferry}} \quad (32)$$

$$= 6000 - 4500 \quad (33)$$

$$= 1500 \text{ kWh} \quad (34)$$

(d) If this trip occurs twice daily for a full year, how much energy is saved annually?

$$\text{Annual energy savings} = 1500 \text{ kWh/trip} \cdot 2 \text{ trips/day} \cdot 365 \text{ days} \quad (35)$$

$$= 1,095,000 \text{ kWh/year} \quad (36)$$

$$= 1095 \text{ MWh/year} \quad (37)$$

(e) If the fuel costs 100 NZD/MWh, how much is the society saving on fuel annually? If the ferry takes 15 min longer than the plane, and if the average person's time is worth 40 NZD/h, how many person-hours are lost in a year, and what is the cost of that time?

Passenger time lost per trip:

$$15 \text{ min} = 0.25 \text{ h}$$

Passengers per trip:

$$200$$

$$\text{person-hours lost per trip} = 200 \cdot 0.25$$

$$= 50 \text{ person-h}$$

Annual person-hours lost:

$$50 \cdot 730$$

$$= 36,500 \text{ person-h/year}$$

Cost of time:

$$36,500 \cdot 40$$

$$= 1,460,000 \text{ NZD/year}$$

$$\text{Fuel saving} = 109,500 \text{ NZD/year}$$

$$\text{Time cost} = 1,460,000 \text{ NZD/year}$$

Question 4

A regional train in the South Island transports on average 180 passengers per trip over a distance of 100 km. The electric train consumes 12 kWh/km. What is the energy consumption in kWh/person·km?

$$\text{Total energy consumption per trip} = 12 \frac{\text{kWh}}{\text{km}} \cdot 100 \text{ km} \quad (38)$$

$$= 1200 \text{ kWh} \quad (39)$$

$$\text{Total person-km} = 180 \text{ persons} \cdot 100 \text{ km} \quad (40)$$

$$= 18,000 \text{ person} \cdot \text{km} \quad (41)$$

$$\text{Energy consumption per person-km} = \frac{1200 \text{ kWh}}{18,000 \text{ person} \cdot \text{km}} \quad (42)$$

$$= 0.0667 \frac{\text{kWh}}{\text{person} \cdot \text{km}} \quad (43)$$

Question 5

A small commuter van (diesel) is used for airport shuttle services. It has a fuel efficiency of 10 km/l and a seating capacity of 10 passengers. The diesel energy content is 10 kWh/l. If the van operates at full capacity over a 200 km trip, calculate the energy consumption per person-km.

$$\text{Fuel used} = \frac{200 \text{ km}}{10 \text{ km/l}} \quad (44)$$

$$= 20 \text{ l} \quad (45)$$

$$\text{Total energy used} = 20 \text{ l} \cdot 10 \frac{\text{kWh}}{\text{l}} \quad (46)$$

$$= 200 \text{ kWh} \quad (47)$$

$$\text{Total person-km} = 10 \text{ persons} \cdot 200 \text{ km} \quad (48)$$

$$= 2000 \text{ person} \cdot \text{km} \quad (49)$$

$$\text{Energy per person-km} = \frac{200 \text{ kWh}}{2000 \text{ person} \cdot \text{km}} \quad (50)$$

$$= 0.1 \frac{\text{kWh}}{\text{person} \cdot \text{km}} \quad (51)$$

Question 6

Christchurch's electric bus fleet carries approximately 35 passengers per trip on average. Assume an electric bus travels 300 km per day and consumes 0.5 kWh/km.

- (a) Calculate the daily person-km transported by one electric bus.

$$\text{Daily person-km} = 35 \text{ persons} \cdot 300 \text{ km} \quad (52)$$

$$= 10,500 \text{ person} \cdot \text{km} \quad (53)$$

- (b) Determine the energy consumption per person-km in kWh.

$$\text{Total daily energy consumption} = 0.5 \frac{\text{kWh}}{\text{km}} \cdot 300 \text{ km} \quad (54)$$

$$= 150 \text{ kWh} \quad (55)$$

$$\text{Energy consumption per person-km} = \frac{150}{10,500} \quad (56)$$

$$= 0.0143 \frac{\text{kWh}}{\text{person} \cdot \text{km}} \quad (57)$$

- (c) Compare this value to a single-occupancy gasoline car that travels 14,000 km/year, with a fuel efficiency of 12 km/l and an energy content of 10 kWh/l. What is the energy consumption per person-km for the car?

$$\text{Fuel consumption per km} = \frac{1}{12} \frac{\text{l}}{\text{km}} \quad (58)$$

$$\text{Energy consumption per km} = \frac{10}{12} \quad (59)$$

$$= 0.833 \frac{\text{kWh}}{\text{km}} \quad (60)$$

Since the car has a single occupant:

$$\text{Energy consumption per person-km} = 0.833 \frac{\text{kWh}}{\text{person} \cdot \text{km}} \quad (61)$$

- (d) Which mode is more energy-efficient per person-km, and by what factor?

$$\text{Efficiency factor} = \frac{0.833}{0.0143} \quad (62)$$

$$\approx 58.3 \quad (63)$$

Therefore, the electric bus is approximately 58 times more energy-efficient per person-km than the single-occupancy gasoline car.

Question 7

A fully loaded electric delivery van accelerates from 30 km/h to 90 km/h in 12 s while merging onto a motorway. The total energy required for this acceleration, including kinetic energy, rolling resistance, and aerodynamic drag, is:

$$E_{\text{total}} = 620 \text{ kJ}$$

Assume the vehicle travels on a flat road with constant acceleration.

The van has:

$$C_{\text{roll}} = 0.012$$

$$C_d = 0.28$$

$$A = 2.8 \text{ m}^2$$

- (a) Calculate the mass of the vehicle. First, convert the initial and final velocities from km/h to m/s:

$$v_1 = 30 \cdot \frac{1000}{3600}$$

$$v_1 = 8.33 \text{ m/s}$$

$$v_2 = 90 \cdot \frac{1000}{3600}$$

$$v_2 = 25.00 \text{ m/s}$$

The acceleration is:

$$a = \frac{v_2 - v_1}{t}$$

$$a = \frac{25.00 - 8.33}{12}$$

$$a = 1.389 \text{ m/s}^2$$

The distance travelled during acceleration is:

$$d = \frac{v_1 + v_2}{2} \cdot t$$

$$d = \frac{8.33 + 25.00}{2} \cdot 12$$

$$d = 200 \text{ m}$$

The total energy is given by:

$$E_{\text{total}} = \Delta KE + W_{\text{roll}} + W_{\text{drag}}$$

The change in kinetic energy is:

$$\Delta KE = \frac{1}{2}m(v_2^2 - v_1^2)$$

Calculate the velocity squared difference:

$$v_2^2 = (25.00)^2 = 625$$

$$v_1^2 = (8.33)^2 = 69.4$$

$$v_2^2 - v_1^2 = 625 - 69.4$$

$$v_2^2 - v_1^2 = 555.6$$

Therefore:

$$\Delta KE = \frac{1}{2}m(555.6)$$

$$\Delta KE = 277.8m$$

The rolling resistance work is:

$$W_{\text{roll}} = C_{\text{roll}} \cdot m \cdot g \cdot d$$

$$W_{\text{roll}} = 0.012 \cdot m \cdot 9.81 \cdot 200$$

$$W_{\text{roll}} = 23.5m$$

The aerodynamic drag work is calculated as:

$$W_{\text{drag}} = \frac{1}{8} \rho C_d A \frac{v_2^4 - v_1^4}{a}$$

Now calculate the fourth powers of the velocities:

$$v_2^4 = (25.00)^4$$

$$v_2^4 = 390,625$$

$$v_1^4 = (8.33)^4$$

$$v_1^4 = 4,820$$

$$v_2^4 - v_1^4 = 390,625 - 4,820$$

$$v_2^4 - v_1^4 = 385,805$$

Now divide by the acceleration:

$$\frac{v_2^4 - v_1^4}{a} = \frac{385,805}{1.389}$$

$$\frac{v_2^4 - v_1^4}{a} = 277,757$$

Now calculate the constant term:

$$\frac{1}{8} \rho C_d A = \frac{1}{8} \cdot 1.225 \cdot 0.28 \cdot 2.8$$

$$\frac{1}{8} \rho C_d A = 0.120$$

Therefore:

$$W_{\text{drag}} = 0.120 \cdot 277,757$$

$$W_{\text{drag}} = 33,331 \text{ J}$$

$$W_{\text{drag}} = 33.3 \text{ kJ}$$

Now substitute all terms into the total energy equation:

$$620,000 = 277.8m + 23.5m + 33,331$$

Combine the mass-dependent terms:

$$620,000 = 301.3m + 33,331$$

Subtract the drag contribution:

$$620,000 - 33,331 = 301.3m$$

$$586,669 = 301.3m$$

Solve for the mass:

$$m = \frac{586,669}{301.3}$$

$$m = 1,947 \text{ kg}$$

Therefore, the vehicle mass is approximately:

$$\boxed{m \approx 1,950 \text{ kg}}$$

(b) Determine the percentage contribution of:

- kinetic energy,
- rolling resistance,
- aerodynamic drag

to the total energy consumption.

Kinetic energy contribution:

$$\Delta KE = 277.8 \cdot 1,947$$

$$\Delta KE = 540,847 \text{ J}$$

$$\Delta KE \approx 541 \text{ kJ}$$

Rolling resistance contribution:

$$W_{\text{roll}} = 23.5 \cdot 1,947$$

$$W_{\text{roll}} = 45,755 \text{ J}$$

$$W_{\text{roll}} \approx 45.8 \text{ kJ}$$

Aerodynamic drag contribution:

$$W_{\text{drag}} \approx 33.3 \text{ kJ}$$

Now calculate the percentage contributions:

Kinetic energy:

$$\frac{541}{620} \cdot 100$$

$$= 87.3\%$$

Rolling resistance:

$$\frac{45.8}{620} \cdot 100$$

$$= 7.4\%$$

Aerodynamic drag:

$$\frac{33.3}{620} \cdot 100$$

$$= 5.4\%$$

- (c) Briefly explain why aerodynamic drag becomes increasingly important at higher vehicle speeds.

Aerodynamic drag becomes increasingly important at higher speeds because the drag force is proportional to the square of velocity:

$$F_{\text{drag}} \propto v^2$$

As vehicle speed increases, aerodynamic losses increase rapidly, requiring significantly more energy to continue accelerating or maintaining speed.